Intermediate Microeconomics

Week 2 · Day 4 — Monopoly (Perloff, Chs. 11–12)

Market power, deadweight loss, and price discrimination

Recap of Day 3, and today's plan

Day 3 built the competitive market from 20 price-taking firms: each set $P = MC$, entry drove profit to zero, and the outcome $(P^*, Q^*) = (30, 60)$ maximised total surplus (1350).

Today we drop assumption 1 — "many sellers"

A single firm now faces the *whole* downward-sloping demand $P = 60 - \frac{1}{2}Q$. To keep the comparison clean, give it the industry's cost: $MC = 15 + \frac{1}{4}Q$ (the old supply curve — one owner, all 20 plants). The competitive benchmark is still $(30, 60)$.

The monopoly problem	why $MR < P$; the profit-maximising (Q_m, P_m)
Welfare	consumer/producer surplus, transfer, deadweight loss
Price discrimination	first-degree (perfect) and third-degree (segmented)

Learning goals — Day 4

1. Define monopoly, name its sources, and state why the firm is a price *maker*.
2. Derive marginal revenue and explain why $MR < P$ for a monopolist.
3. Solve the monopoly problem $MR = MC$ and compare (Q_m, P_m) to the competitive outcome.
4. Measure the welfare cost: the surplus transfer and the deadweight loss, via the Lerner index and elasticity.
5. Analyse first-degree price discrimination (efficient quantity, zero consumer surplus).
6. Analyse third-degree price discrimination and the inverse-elasticity pricing rule.

What is a monopoly?

A **monopoly** is a single seller of a product with no close substitutes. Unlike the price taker, it faces the entire market demand curve, so raising its price loses *some* — not all — of its buyers. It is a **price maker**.

Where monopoly (or near-monopoly) shows up

- **Utilities** — local water and electricity distribution (natural monopolies, usually regulated).
- **Patents** — a pharmaceutical firm holds a legal monopoly on a drug for the patent term.
- **Control of a resource** — De Beers historically dominated rough diamonds.
- **Network effects / platforms** — a dominant operating system or search engine (market definition permitting).

Pure monopoly is rare; the model matters because *market power* — price above marginal cost — is everywhere.

Where does the power come from? Barriers to entry

Monopoly survives only if entry is blocked. The classic barriers:

1. **Economies of scale** — if average cost falls over the whole relevant range, one firm serves the market more cheaply than many: a *natural monopoly*.
2. **Legal barriers** — patents, copyrights, licences, government franchises.
3. **Control of an essential input** — own the only mine, own the market.
4. **Network effects** — the product is more valuable the more people use it, so an incumbent is hard to dislodge.

Without a barrier, Day 3's logic returns: profit attracts entry, and the monopoly erodes. The barrier is what lets $P > MC$ persist.

Marginal revenue: why $MR < P$

The monopolist sells more only by lowering price — on *every* unit, not just the last. So marginal revenue falls short of price.

With $P(Q) = 60 - \frac{1}{2}Q$,

$$TR = P(Q)\,Q = 60Q - \tfrac{1}{2}Q^2,$$

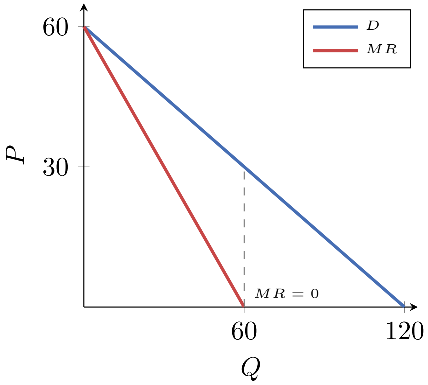
$$MR = \frac{\mathrm{d}TR}{\mathrm{d}Q} = 60 - Q.$$

Same intercept as demand, *twice* the slope.

The general identity

$$MR = P\left(1 + \tfrac{1}{\varepsilon}\right).$$

Since $\varepsilon < 0$, $MR < P$; and $MR > 0$ only where demand is **elastic** ($|\varepsilon| > 1$). A monopolist never operates on the inelastic part of demand.



MR hits zero at the unit-elastic midpoint.

Profit maximisation: $MR = MC$

Maximise $\pi(Q) = P(Q)Q - C(Q)$; FOC $MR = MC$.

Step 1 — set $MR = MC$

$$60 - Q = 15 + \tfrac{1}{4}Q.$$

Step 2 — solve for Q_m

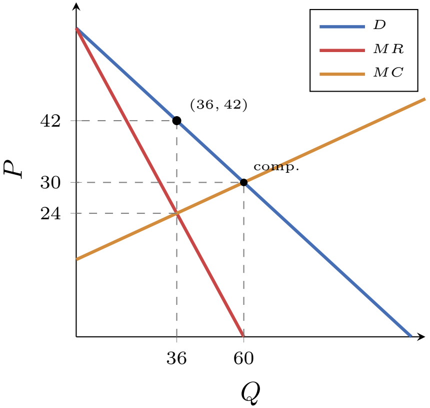
$$45 = \tfrac{5}{4}Q \Rightarrow Q_m = 36.$$

Step 3 — price off the demand curve

$$P_m = 60 - \tfrac{1}{2}(36) = 42.$$

Numeric check: $MR = 60 - 36 = 24 = 15 + 9 = MC$. ✓

Monopoly cuts output $(36 < 60)$ *and raises price* $(42 > 30)$.



The markup: Lerner index and elasticity

The monopolist's price–cost margin is summarised by the **Lerner index**:

$$L = \frac{P - MC}{P} = -\frac{1}{\varepsilon}.$$

At our optimum, using $\varepsilon = \dfrac{\mathrm{d}Q}{\mathrm{d}P}\dfrac{P}{Q} = (-2)\dfrac{42}{36} = -\dfrac{7}{3}$:

$$L = \frac{42 - 24}{42} = \frac{18}{42} = \frac{3}{7} \approx 0.43, \qquad -\frac{1}{\varepsilon} = \frac{3}{7}. \checkmark$$

Reading

Markup is *inversely* proportional to elasticity: the less price-sensitive the buyers, the higher the markup. A price taker faces $\varepsilon = -\infty$, giving $L = 0$ ($P = MC$) — the competitive limit.

Consumer surplus, producer surplus, deadweight loss

By restricting output to 36 and pricing at 42:

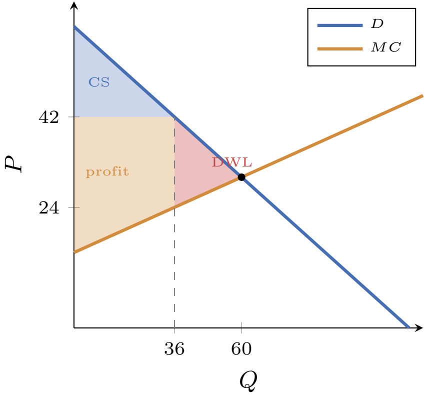
$$CS_m = \tfrac{1}{2}(36)(60-42) = 324,$$

$$PS_m = \int_0^{36} (42 - MC)\,\mathrm{d}Q = 810,$$

$$TS_m = 1134, \qquad DWL = 1350 - 1134 = 216.$$

The transfer, made explicit

Relative to competition: consumers lose $900 - 324 = \mathbf{576}$; the monopolist gains $810 - 450 = \mathbf{360}$; the wedge $576 - 360 = \mathbf{216}$ is destroyed — nobody's.



First-degree (perfect) price discrimination

Suppose the monopolist knows every buyer's willingness to pay and charges exactly that. Then price *equals* the demand curve for each unit, so $MR =$ demand, and it produces until demand meets MC.

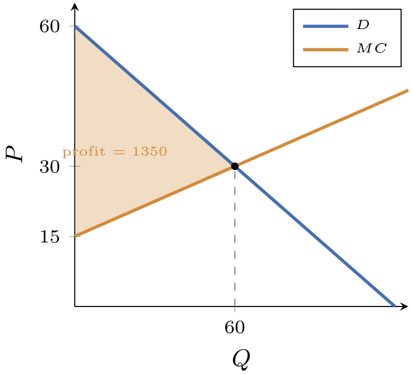
Efficient quantity

$$60 - \tfrac{1}{2}Q = 15 + \tfrac{1}{4}Q \Rightarrow Q = 60.$$

Same as competition — **no deadweight loss.**

But all surplus is captured

$$\pi = \int_0^{60} \big(P(Q) - MC\big)\mathrm{d}Q = 1350 = TS, \quad CS = 0.$$



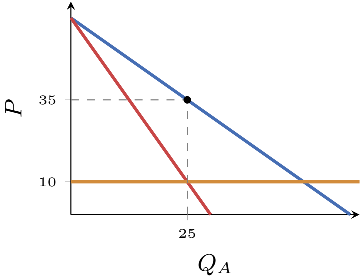
The paradox

Perfect discrimination is *efficient* (produces 60) yet maximally *unfair* (consumers keep nothing). Efficiency and distribution are separate questions.

Third-degree price discrimination: segment the market

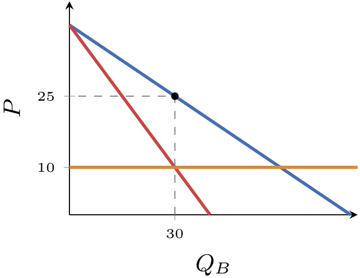
Charge different prices to separable groups (students vs. adults, home vs. export). The rule: set $MR = MC$ *in each market*. With constant $MC = 10$ and two groups:

Market A (less elastic)



$60 - 2Q_A = 10 \Rightarrow Q_A = 25,\ P_A = 35$

Market B (more elastic)



$40 - Q_B = 10 \Rightarrow Q_B = 30,\ P_B = 25$

Elasticities at the optima: $\varepsilon_A = -\frac{7}{5}$, $\varepsilon_B = -\frac{5}{3}$. The **less elastic market pays more** ($P_A = 35 > P_B = 25$) — inverse-elasticity pricing, $\dfrac{P_i - MC}{P_i} = -\dfrac{1}{\varepsilon_i}$.

Day 4 checkpoint

You can now

- derive $MR = 60 - Q$ and explain why $MR < P$ (and why monopoly avoids the inelastic region);
- solve $MR = MC$: monopoly $(36, 42)$ vs competition $(60, 30)$;
- decompose the welfare loss — transfer 576 from consumers, 360 to the firm, $DWL = 216$;
- use the Lerner index $L = -1/\varepsilon = 3/7$;
- contrast first-degree PD (efficient, $CS = 0$) with third-degree PD (higher price to the less elastic group).

The week, closed

supply and demand (Day 1) → welfare and taxes (Day 2) → the competitive firm (Day 3) → its opposite, monopoly (Day 4). The same market ran through all four.

In-class exercises — Day 4

Time for the in-class exercises.

- **Exercise 1** — single-price monopoly and then first-degree discrimination on the same market: quantity, price, surplus, and deadweight loss.
- **Exercise 2** — third-degree discrimination across two markets: the price in each, and the elasticity rule behind the gap.

Solutions are at the end of the exercise sheet — try both fully before turning there.